

SymPy Bug Card

Bug 49: Matrix.solve Methods Return Malformed Results

Task. Solve a singular but consistent linear system with explicit *SymPy* matrix-solver methods.

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad A u = b.$$

SymPy's answer. `A.solve(b, method="QR") = [1]`, while `method="LDL"` and `method="CRAMER"` return $\begin{bmatrix} \text{nan} \\ \text{nan} \end{bmatrix}$.

Correct answer. The system is non-unique:

$$u = \begin{bmatrix} 1 - t \\ t \end{bmatrix}, \quad t \in \mathbb{C},$$

so `Matrix.solve` should reject the solve or return a documented full solution representation.

Explanation. The second equation is twice the first, so $x + y = 1$ is the only independent constraint. A one-entry vector cannot solve a two-unknown system, and substituting the nan vector gives nan residuals rather than 0.

Likely root cause. The diagnosed dispatch path in `sympy/matrices/solvers.py` applies the non-unique-solution guard to the default GJ/GE path but sends QR, LDL, and CRAMER directly to method-specific solvers. Those branches then return a reduced-shape result or divide by zero, making this a new failure mode with no public precedent.

Artifact (GitHub).

[bug-report/bug-049-matrix-solve-methods-return-malformed-results/](#)